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REMOTE SUPPORT DEVICE, REMOTE SUPPORT METHOD AND COMPUTER-READABLE MEDIUM

Abstract

An image delay compensation time α is set to be equal to or less than an image delay time indicating a timing difference D1 between the timing T1 at which an image IMG1 is acquired by a camera CAM and a timing T3 at which a remote operator terminal 200 decodes the image IMG1. Based on information on a movement of a mobile vehicle, projection transformation is performed in which the image IMG1 is transformed into a future image IMG2 obtained at a camera viewpoint at a timing T2 that is later than the timing T1 by the image delay compensation time α . A synthetic image SIMG1 is generated based on the future image IMG2 and recognition information OR on the object in the image IMG1 and is output from a display device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-035759, filed on Mar. 8, 2024, the contents of which application are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a technique for remotely supporting a mobile vehicle using an image acquired by a camera mounted on the mobile vehicle.

BACKGROUND

[0003] JP2022159912A discloses a system for remotely monitoring a vehicle. In this system of the related art, first data and second data are separately transmitted from the vehicle to a remote server. The first data includes an image (a camera image) acquired by a camera mounted on the vehicle and acquisition timing of the camera image. The second data includes an object recognition result based on the camera image and an acquisition timing of the camera image. In the remote server, based on the acquisition timing of the camera image included in the first and second data, the camera image acquired at the same timing and the object recognition result based on the camera image are extracted in time series. Then, a synthetic image in which the object recognition result is superimposed on the extracted camera image is output from a remote monitoring screen.

[0004] In the system of the related art, a difference between the acquisition timing of the camera image and a timing at which the camera image can be processed by the remote server can be said to be a delay time of the camera image. The delay time of the camera image includes a time required for communication between the vehicle and the remote server and a time required for processing executed in the vehicle. The delay time corresponding to the delay time of the camera image is also present in the object recognition result. The delay time of the object recognition result can be said to be a difference between the acquisition timing of the camera image that is the source of the object recognition result and the timing at which the object recognition result is able to be processed by the remote server.

[0005] The problem here is that the timing at which the camera image is able to be processed by the remote server and the timing at which the object recognition result based on the camera image is able to be processed by the remote server are not the same. The fact that these timings are not aligned means that after one of the pieces of information becomes processable, it needs to wait until the other information becomes processable. This leads to a delay in the output of the synthetic image from the remote monitoring screen.

[0006] The present disclosure has been made in view of the above problems. An object of the present disclosure is to provide a technique for suppressing the delay in the output of the synthetic image in which the object recognition result is superimposed on the camera image from the remote support screen, when the camera image and the object recognition result are separately received from the mobile vehicle and the remote support of the mobile vehicle is performed.

SUMMARY

[0007] A first aspect of the present disclosure is a device for remotely supporting a mobile vehicle and has the following features.

[0008] The device is configured to remotely support the mobile vehicle by outputting a synthetic image in which annotation information about an object to be noted in an image is superimposed on the image obtained by a camera mounted on the mobile vehicle from a display device.

[0009] The device includes a communication circuitry and a processing circuitry. The

communication circuitry is connected to the mobile vehicle via a communication network. The processing circuitry is coupled to the communication circuitry.

[0010] The communication circuitry is configured to: receive, from the mobile vehicle, image data obtained by encoding the image and information on acquisition timing of the image by the camera; and receive, from the mobile vehicle, object data including recognition information on an object in the image separately from the image data.

[0011] The processing circuitry is configured to: set an image delay compensation time equal to or less than an image delay time indicating a timing difference between an acquisition timing of the image included in the image data and a decoding timing of the image data by the processing circuitry; perform projection transformation of the image included in the image data into a future image obtained at a camera viewpoint earlier than the acquisition timing of the image by the image delay compensation time based on information regarding a movement of the mobile vehicle; and generate the synthetic image based on the future image and recognition information on the object in an original image of the future image.

[0012] A second aspect of the present disclosure is a method for remotely supporting a mobile vehicle and has the following features.

[0013] The method includes remotely supporting the mobile vehicle by outputting, from a display device, a synthetic image in which annotation information about an object to be noted in an image is superimposed on the image obtained by a camera mounted in the mobile vehicle.

[0014] The method includes: receiving image data in which the image and information on acquisition timing of the image by the camera are encoded from the mobile vehicle; receiving object data including recognition information on an object in the image from the mobile vehicle separately from the image data; setting an image delay compensation time equal to or less than an image delay time indicating a timing difference between a decoding timing of the image data and an acquisition timing of the image included in the image data; performing projection transformation of the image included in the image data into a future image obtained at a camera viewpoint earlier than the acquisition timing of the image by the image delay compensation time based on information regarding a movement of the mobile vehicle; and generating the synthetic image based on the future image and recognition information on the object in an original image of the future image.

[0015] A third aspect of the present disclosure is a non-transitory computer-readable medium storing a program for remotely supporting a mobile vehicle and has the following features.

[0016] The program is configured to cause a computer to function to remotely support the mobile vehicle by outputting, from a display device, a synthetic image in which annotation information about an object to be noted in an image obtained by a camera mounted on the mobile vehicle is superimposed on the image.

[0017] The program is configured to cause the computer to execute: receiving image data in which the image and information on acquisition timing of the image by the camera are encoded from the mobile vehicle; receiving object data including recognition information on an object in the image from the mobile vehicle separately from the image data; setting an image delay compensation time equal to or less than an image delay time indicating a timing difference between a decode timing of the image data and an acquisition timing of the image included in the image data; projection transforming the image included in the image data into a future image obtained at a camera viewpoint earlier than the acquisition timing of the image by the image delay compensation time based on information regarding the movement of the mobile vehicle; and generating the synthetic image based on the future image and recognition information on the object in an original image of the future image.

[0018] According to the present disclosure, the projection transformation of the image is performed in the processing circuitry. According to the projection transformation of the image, the future image obtained at a camera viewpoint earlier than the acquisition timing of the image by the image

delay compensation time is obtained. Therefore, for example, before the image acquired by the camera at a timing TA can be processed in the processing circuitry, the future image can be obtained based on the image acquired by the camera at a timing TB before the timing TA, and the synthetic image can be generated based on the future image. Therefore, it is possible to suppress the delay of the output of the synthetic image from the display device.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0019] FIG. 1 is a schematic diagram illustrating a configuration example of a remote support system;

[0020] FIG. 2 is a diagram illustrating an example of data transmission from a vehicle to a remote operator terminal;

[0021] FIG. 3 is a diagram illustrating an example of a synthetic image output from a display device;

[0022] FIG. 4 is a diagram for illustrating a delay time of an image and a delay time of an object;

[0023] FIG. 5 is a conceptual diagram for illustrating an outline of delay compensation processing of an image;

[0024] FIG. 6 is a conceptual diagram for illustrating projection transformation based on perspective projection transformation;

[0025] FIG. 7 is a diagram illustrating an example of a synthetic image output from the display device when the projection transformation is performed on an image;

[0026] FIG. 8 is a conceptual diagram for illustrating an outline of movement adjustment processing of an object;

[0027] FIG. 9 is a diagram illustrating an example of a synthetic image output from a display device when the projection transformation is performed on a partial image of an object;

[0028] FIG. 10 is a block diagram illustrating a configuration example of a vehicle; and

[0029] FIG. 11 is a block diagram illustrating an example of a configuration of a remote operator terminal.

DESCRIPTION OF EMBODIMENT

[0030] An embodiment of the present disclosure will be described with reference to the accompanying drawings.

1. Remote Support System

[0031] A remote support (a remote driving) of a mobile vehicle will be considered. Examples of the mobile vehicle to be remotely supported include a vehicle, a robot, and a flying object. The vehicle may be an autonomous driving vehicle, or a vehicle driven by a driver. Examples of the robot include a logistics robot and a work robot. Examples of the flying object include a drone or the like. As an example, in the following description, a case where the mobile vehicle is a vehicle will be considered. In the case of generalization, “vehicle” in the following description is replaced with “mobile vehicle”.

[0032] FIG. 1 is a schematic diagram illustrating a configuration example of a remote support system 1 according to the present embodiment. The remote support system 1 includes a vehicle 100, a remote operator terminal 200, and a management device 300. The vehicle 100 is a target of the remote support. The remote operator terminal 200 is a terminal device used when a remote operator O remotely supports the vehicle 100. The remote operator terminal 200 may be referred to as a remote support human machine interface (HMI). The management device 300 manages the remote support system 1. Typically, the management device 300 is a management server on a cloud. The management server may be configured by a plurality of servers that perform distributed processing.

[0033] The vehicle **100**, the remote operator terminal **200**, and the management device **300** can communicate with each other via a communication network. The vehicle **100** and the remote operator terminal **200** may communicate with each other through the management device **300**. The vehicle **100** and the remote operator terminal **200** may directly communicate with each other without the management device **300**.

[0034] Various sensors including a camera CAM are mounted on the vehicle **100**. The camera CAM captures an image of surroundings of the vehicle **100** and acquires an image IMG showing a situation around the vehicle **100**. Information on sensor detection SEN includes information obtained by the various sensors. The information on sensor detection SEN includes at least the image IMG captured by the camera CAM and recognition information OR on an object to be noted in the image IMG. The information on sensor detection SEN may include a position and a state (e.g., speed, steering angle, etc.) of the vehicle **100**. The vehicle **100** transmits information on sensor detection SEN to the remote operator terminal **200**.

[0035] The remote operator terminal **200** receives the information on sensor detection SEN transmitted from the vehicle **100**. The remote operator terminal **200** presents the information on sensor detection SEN to the remote operator O. Specifically, the remote operator terminal **200** includes a display device **220**, and displays information such as the image IMG on the display device **220**. The remote operator O recognizes the situation around the vehicle **100** by viewing the displayed information, and performs the remote support of the vehicle **100**. That is, the remote support of the vehicle **100** by the remote operator O is performed by displaying the information on the remote operator O on the display device **220**.

[0036] Remote support information OPE is information on the remote support by the remote operator O. For example, the remote support information OPE includes an operation amount by the remote operator O. The remote operator terminal **200** transmits the remote support information OPE to the vehicle **100**. The vehicle **100** receives the remote support information OPE transmitted from the remote operator terminal **200**. The vehicle **100** performs vehicle travel control in accordance with the received remote support information OPE. In this way, the remote support of the vehicle **100** is realized.

2. Generation Processing of Synthetic Image

[0037] In the present embodiment, a synthetic image SIMG is generated in the remote operator terminal **200**. The synthetic image SIMG is generated based on the image IMG included in the information on sensor detection SEN and the recognition information OR on the object to be noted in the image IMG. The image IMG and the recognition information OR are separately transmitted from the vehicle **100** to the remote operator terminal **200**.

[0038] FIG. **2** is a diagram illustrating an example of transmission of the image IMG and the recognition information OR from the vehicle **100** to the remote operator terminal **200**. Image data including a set of images IMG (i.e., video) acquired within a certain time is transmitted from the vehicle **100** to the remote operator terminal **200**. The certain time corresponds to a transmission interval of the image data. In the example shown in FIG. **2**, the set of images IMG1 acquired at the timing T1 includes images IMG1 (T1a), IMG1 (T1b), and IMG1 (T1c). The images IMG1 (T1a), IMG1 (T1b) and IMG1 (T1c) are acquired at the timings T1a, T1b, and T1c, respectively.

[0039] The recognition processing of the object is performed on each of the images IMG1 (T1a), IMG1 (T1b) and IMG1 (T1c). For the recognition processing, an image analysis technique such as pattern matching or deep learning is used. When the recognition processing is performed, recognition information OR (T1a), OR (T1b) and OR (T1c) are generated. The recognition information OR includes timing (e.g., a time stamp) at which the image IMG1 subjected to the recognition processing is acquired by the camera CAM. When an object to be noted is recognized in the image IMG1, coordinates, size, and type of the object in the image IMG1 are added to the recognition information OR. That is, when the object to be noted is not recognized in the image IMG1, the recognition information OR includes only the timing information (the time stamp).

[0040] The object data including the recognition information OR is transmitted from the vehicle **100** to the remote operator terminal **200**. The transmission of the object data is performed each time the recognition information OR is generated. The transmission of the object data may be performed at regular time intervals. The regular time corresponds to a transmission interval of the object data.

[0041] The remote operator terminal **200** that has received the image data, and the object data arranges the images IMG1 (i.e., the images IMG1 (T1a), IMG1 (T1b), and IMG1 (T1c)) in chronological order based on the timing (the time stamp) at which the images IMG1 included in the image data are acquired by the camera CAM. The remote operator terminal **200** also specifies an image IMG1 to be combined with the recognition information OR based on the timing (the time stamp) included in the recognition information OR on the object.

[0042] The remote operator terminal **200** further refers to the coordinates, size, and type of the object included in the recognition information OR on the object, and superimposes annotation information OA on the image IMG1 to be combined with the object. The annotation information OA is information indicating the position, size, and type of the object in the image IMG1. The information indicating the position and size of the object is a bounding box surrounding the object. The information indicating the type of the object is a character indicating the type.

[0043] The annotation information OA is superimposed on the image IMG1, and thus the synthetic image SIMG1 is generated. Since the original images IMG1 (i.e., the images IMG1 (T1a), IMG1 (T1b), and IMG1 (T1c)) of the synthetic image SIMG1 are arranged in chronological order, a set of the synthetic image SIMG1 (i.e., the video) is output from the display device **220** by outputting the synthetic image in chronological order.

[0044] FIG. 3 is a diagram showing an example of the synthetic image SIMG output from the display device **220**. In the example shown in FIG. 3, annotation information OA1 and OA2 are output. The annotation information OA1 is a bounding box surrounding a green light and characters “green light”. The annotation information 2 is a bounding box surrounding a walker and characters “walker”.

3. Processing using Projection Transformation

3-1. Delay Compensation Processing of Image

[0045] The remote support of the vehicle **100** is accompanied by a delay between the vehicle **100** and the remote operator terminal **200**. A difference between the timing T1 at which the image IMG1 is acquired by the camera CAM and the timing at which the image IMG1 is able to be processed by the remote operator terminal **200** can be said to be a delay time of the image. Further, a difference between the timing T1 at which the image IMG1 in which the object to be noted is recognized is acquired by the camera CAM and the timing at which the recognition information OR is able to be processed by the remote operator terminal **200** can be said to be a delay time of the object.

[0046] FIG. 4 is a diagram for illustrating the delay time of the image and the delay time of the object. In the example shown in FIG. 4, after the image IMG1 is acquired at the timing T1, it is encoded and then sent to the remote operator terminal **200**. The reason for encoding the image is to reduce the communication cost. In order for the encoded image IMG1 to be processed by the remote operator terminal **200**, the image IMG1 needs to be decoded. Therefore, the delay time of the image is represented as a timing difference $D1(=T3-T1)$ between a timing T3 at which the remote operator terminal **200** decodes the image IMG1 and the timing T1.

[0047] On the other hand, the recognition information OR is transmitted to the remote operator terminal **200** after being generated. Therefore, the recognition information OR is able to be processed by the remote operator terminal **200** at a timing T4 when the remote operator terminal **200** receives the recognition information OR. Therefore, the delay time of the object is represented as a timing difference $D2(=T4-T1)$ between the timing T4 and the timing T1.

[0048] To generate the synthetic image SIMG1, the image IMG1 and recognition information OR on the image IMG1 are required. However, the timing T3 at which the image IMG1 is decoded is

later than the timing **T4** at which the recognition information OR is received. Therefore, the processing of the recognition information OR has to wait for the decoding of the image **IMG1** for a duration corresponding to a timing difference $D3(T4-T3)$ between the timing **T3** and the timing **T4**.

[0049] Therefore, in the present embodiment, a delay compensation is visually performed on the image **IMG1** in consideration of the delay time (i.e., the timing difference **D1**) of the image **IMG1**. In particular, the remote support system **1** according to the present embodiment visually performs the delay compensation on the image **IMG1** using a “projection transformation”. An entity of the delay compensation processing is, for example, the remote operator terminal **200**. However, the entity of the delay compensation processing is not limited to the remote operator terminal **200**. That is, at least a part of delay compensation processing may be executed by the vehicle **100** or the management device **300**.

[0050] FIG. 5 is a conceptual diagram for illustrating an outline of the delay compensation processing of an image by the remote support system **1**. The image **IMG1** is an image **IMG** actually captured at the timing **T1** by the camera **CAM** mounted on the vehicle **100**. The image **IMG1** is transmitted from the vehicle **100** to the remote operator terminal **200**. The remote operator terminal **200** acquires the image **IMG1** after the timing **T1**. If an image **IMG** to be captured in the future can be predicted from the image **IMG1**, the delay compensation can be performed.

[0051] The timing **T2** is a timing that is a target of prediction, and is a timing after the timing **T1**. The difference between the timing **T2** and the timing **T1** corresponds to the “delay compensation time”. The remote support system **1** sets a time α equal to or less than the timing difference **D1** shown in FIG. 4 as a delay compensation time of the image (an image delay compensation time). In the first example, the time α is set to a time equal to the timing difference **D1**. According to the first example, the synthetic image **SIMG1** can be generated after the timing (e.g., the timing **T4**) at which the transmission of the recognition information OR is completed. That is, the timing difference $D3(=T4-T3)$ described in FIG. 4 can be eliminated, and the synthetic image **SIMG1** can be output at an earlier timing.

[0052] In the second example, the time α is set to a time that is equal to or longer than the timing difference **D2** and shorter than the timing difference **D1**. According to the second example, it is possible to shorten the timing difference $D3(=T4-T3)$ described in FIG. 4. In the third example, the time α is set to a time that is equal to or longer than the timing difference **D3** and shorter than the timing difference **D2**. According to the third example, the synthetic image **SIMG1** can be generated at a timing after a timing at which the transmission of the recognition information OR is completed (i.e., the timing **T4**) and before the timing **T3**. In any case, the remote support system **1** sets the timing **T2** to be later than the timing **T1** by the image delay compensation time.

[0053] For convenience, the camera **CAM** at the timing **T1** is referred to as a first camera **CAM1**, and the camera **CAM** at the timing **T2** is referred to as a second camera **CAM2**. A first viewpoint is a viewpoint of the first camera **CAM1** and is defined by a combination of the position and the direction of the first camera **CAM1** at the timing **T1**. A second viewpoint is a viewpoint of the second camera **CAM2** and is defined by a combination of the position and the direction of the second camera **CAM2** predicted at the timing **T2**.

[0054] The remote support system **1** acquires camera information **CINF** regarding the camera **CAM** mounted on the vehicle **100**. The camera information **CINF** includes installation information and performance information on the camera **CAM**. The installation information indicates an installation position and an installation orientation of the camera **CAM** in a vehicle coordinate system. The performance information indicates a focal length, an angle of view, and the like of the camera **CAM**. Since the camera **CAM** is fixed to the vehicle **100**, a movement direction and a movement amount of the vehicle **100** can be converted into the movement direction and the movement amount of the camera **CAM** in the camera coordinate system by using the installation information on the camera **CAM**. That is, a change in viewpoints of the camera **CAM** can be

estimated based on the installation information on the camera CAM and the movement direction and the movement amount of the vehicle **100**.

[0055] More specifically, the remote support system **1** acquires information regarding a movement of the vehicle **100** (i.e., the movement direction and the movement amount) in a period from the timing T1 to the timing T2 (i.e., the image delay compensation time). For example, the remote support system **1** estimates the movement direction and the movement amount of the vehicle **100** in the period from the timing T1 to the timing T2, based on speed and steering angle of the vehicle **100** at the timing T1 and the image delay compensation time. Information on the speed and steering angle of the vehicle **100** may be obtained from information on sensor detection SEN provided from the vehicle **100**. Alternatively, the steering angle in the steering operation by the remote operator O may be regarded as the steering angle of the vehicle **100**. It may be assumed that the vehicle **100** makes a steady circular turn. Then, the remote support system **1** calculates a difference between the first viewpoint and the second viewpoint based on the camera information CINF (the installation information) and the movement amount and the movement direction of the vehicle **100** in the image delay compensation time.

[0056] The image IMG1 can be said to be an image IMG captured from the first viewpoint, that is, an image IMG viewed from the first viewpoint. An image IMG expected to be captured from the second viewpoint, that is, the image IMG expected to be viewed from the second viewpoint is hereinafter referred to as an “image IMG2”. The remote support system **1** converts the image IMG1 viewed from the first viewpoint into the image IMG2 viewed from the second viewpoint based on the difference between the first viewpoint and the second viewpoint. In other words, the remote support system **1** predicts the image IMG2 viewed from the second viewpoint based on the image IMG1 viewed from the first viewpoint. Projection transformation is used for this prediction.

[0057] FIG. 6 is a conceptual diagram for illustrating the projection transformation. The projection transformation is performed based on the perspective projection transformation. The perspective projection transformation is a rendering method for drawing an object in a three-dimensional space on a two-dimensional plane as viewed from the camera CAM. For that purpose, the perspective projection transformation projects the points in the three-dimensional space onto the projection plane P, taking into account the viewpoint of the camera CAM. The projection plane P is associated with the camera CAM. For example, the projection plane P is a plane orthogonal to the optical axis of the camera CAM. Note that a point in the three-dimensional space is defined in a three-dimensional world coordinate system (absolute coordinate system). On the other hand, the point projected on the projection plane P is defined in a two-dimensional image coordinate system.

[0058] For example, N virtual points are virtually set in the three-dimensional world coordinate system. N is an integer of 4 or more. The N virtual points when viewed from the first camera CAM1 (the first viewpoint) are projected to a first projection PI associated with the first camera CAM1 by the perspective projection transformation. In addition, the N virtual points when viewed from the second CAM2 (the second viewpoint) are projected to a second projection plane P2 associated with the second camera CAM2 by the perspective projection transformation. The second viewpoint is obtained from the difference between the first viewpoint and the second viewpoint. Image coordinates of a virtual point on the first projection plane P1 as viewed from the first camera CAM1 (the first viewpoint) are given by $[x, y]$. On the other hand, image coordinates of a virtual point on the second projection plane P2 as viewed from the second camera CAM2 (the second viewpoint) are given by $[x', y']$. Based on the comparison between the two image coordinates, a projection transformation matrix H for transforming the first viewpoint to the second viewpoint is calculated. Then, the projection transformation matrix H is applied to an entire of the image CAM1 actually captured by the first camera IMG1, thereby generating the image IMG2 expected to be viewed from the second viewpoint.

[0059] As another example, a method described in a non-patent literature (Koudai Matsubara and Manabu Omae, “Compensation of Camera Image Latency in Remotely Operated Vehicle using

Projection Transformation”, 19th ITS Symposium 2021,4-A-12, December 2021) may be used. In particular, each image coordinate point on an image IMG1 (a projection plane P) is transformed into a world coordinate point in the world coordinate system by an inverse transformation of the perspective projection transformation. Based on the difference between the first viewpoint and the second viewpoint, the world coordinate point viewed from the first viewpoint is converted into the world coordinate point viewed from the second viewpoint. Then, the world coordinate point viewed from the second viewpoint is returned onto the projection plane P by perspective projection transformation. Thus, an image IMG2 expected to be viewed from the second viewpoint is generated. According to the non-patent literature, it is assumed that what is reflected in the entire image IMG is the ground surface S as shown in FIG. 6.

[0060] FIG. 7 is a diagram showing an example of a synthetic image SIMG output from the display device 220 when the projection transformation is performed on an image. Each of the annotation information OA1 and OA2 shown in FIG. 7 is the same as those described in FIG. 3. What is important in the description of FIG. 7 is that the synthetic image SIMG2 is generated based on the image IMG2. The synthetic image SIMG2 can be generated before a timing at which the image IMG1 is decoded (i.e., the timing T3 shown in FIG. 4). Therefore, the synthetic image SIMG2 is output from the display device 220, and thus the delay in the output of the synthetic image SIMG due to the delay time of the image is compensated.

3-2. Movement Adjustment Processing of a Moving Body

[0061] The section 3-1. is based on the delay time of the image. In this section, the delay time (the timing difference D2 shown in FIG. 4) of the object is focused. If the delay compensation processing of the image is performed, a future image IMG2 can be generated. However, in the case where the object is a moving body such as a walker, a bicycle, or other vehicles, when the movement amount within the delay time of the object of the moving body is large, the position and size of the moving body in the image IMG2 may deviate greatly from the actual position and size of the moving body. Therefore, in the present embodiment, the “projection transformation” is also performed on the image of the moving body included in the image IMG1.

[0062] FIG. 8 is a conceptual diagram for illustrating an outline of the movement adjustment processing of the moving body by the remote support system 1. The image IMG1 is an image IMG actually captured at the timing T1 by the camera CAM mounted on the vehicle 100. The image IMG1 is transmitted from the vehicle 100 to the remote operator terminal 200. The remote operator terminal 200 extracts a partial image OBJ1 of the moving body from the image IMG1 acquired after the timing T1 based on the recognition information OR. Then, a partial image OBJ2 of the moving body to be captured in the future is predicted from the partial image OBJ1 using the projection transformation.

[0063] The timing T2 shown in FIG. 8 is a timing that is a target of the prediction and is a timing after the timing T1. The difference between the timing T2 and the timing T1 corresponds to the “movement adjustment time”. The remote support system 1 sets a time β equal to or less than the timing difference D2 shown in FIG. 4 as the movement adjustment time of the moving body (the object movement adjustment time). In the first example, the time β is set to a time equal to the timing difference D2. According to the first example, the position and size of the moving body at the time corresponding to the timing difference D2 can be made closer to the actual position and size.

[0064] In the second example, the time β is set to a time less than the timing difference D2. Even in the second example, it is possible to suppress the position and size of the moving body at the time corresponding to the timing difference D2 from greatly deviating from the actual position and size. In any case, the remote support system 1 sets the timing T2 to be later than the timing T1 by the object movement adjustment time.

[0065] The remote support system 1 acquires the camera information CINF regarding the camera CAM mounted on the vehicle 100. Then, a change in viewpoints of the camera CAM is estimated

based on the installation information on the camera CAM included in the camera information CINF. The processing up to this point is the same as the video delay compensation processing described with reference to FIG. 5.

[0066] In the movement adjustment processing of the object, information regarding the relative movement of the moving body with respect to the vehicle **100** (the relative movement direction and relative movement amount) in a period from the timing T1 to the timing T2 (i.e., the object movement adjustment time) is acquired. For example, the remote support system **1** estimates the relative movement direction and the relative movement amount of the moving body from the timing T1 to the timing T2 by an optical flow. Then, the remote support system **1** calculates a difference between the first viewpoint and the second viewpoint based on the camera information CINF (the installation information) and the relative movement amount and the relative movement direction of the moving body in the object movement adjustment time. Based on the difference, the remote support system **1** converts the image IMG1 viewed from the first viewpoint into the image IMG2 viewed from the second viewpoint.

[0067] FIG. 9 is a diagram showing an example of a synthetic image SIMG output from the display device **220** when projection transformation is performed on a partial image of the moving body. Each of the annotation information OA1 and OA2 shown in FIG. 9 is the same as those described in FIGS. 3 and 7. What is important in the description of FIG. 9 is that the position and size (shown with a solid line) of the annotation information OA2 (walker) are different from the position and size (shown with a broken line) of the annotation information OA2 in FIG. 7. This is because the projection transformation is performed on the partial image of the moving body.

4. Configuration Example of Vehicle

4-1. Configuration Example

[0068] FIG. 10 is a block diagram showing a configuration example of the vehicle **100**. The vehicle **100** includes a communication device **110**, sensors **120**, a driving device **130**, and a control device **150**.

[0069] The communication device **110** communicates with the outside of the vehicle **100**. For example, the communication device **110** communicates with the remote operator terminal **200** or the management device **300**.

[0070] The sensors **120** include a recognition sensor, a vehicle state sensor, a position sensor, and the like. The recognition sensor recognizes (detects) a situation around the vehicle **100**. Examples of the recognition sensor include the camera CAM, a laser imaging detection and ranging (LIDAR), and a radar. The vehicle state sensor detects a state of the vehicle **100**. The vehicle state sensor includes a speed sensor, an acceleration sensor, a yaw rate sensor, a steering angle sensor, and the like. The position sensor detects the position and the orientation of the vehicle **100**. For example, the position sensor includes a GNSS sensor.

[0071] The driving device **130** includes a steering device, a driving device, and a braking device. The steering device steers the wheels. For example, the steering device includes an electric power steering (EPS) device. The driving device is a power source that generates a driving force. Examples of the driving device include an engine, an electric motor, and an in-wheel motor. The braking device generates a braking force.

[0072] The control device **150** is a computer that controls the vehicle **100**. The control device **150** includes one or more processors **160** (hereinafter, simply referred to as a “processor **160**”) and one or more memory devices **170** (hereinafter, simply referred to as a “memory device **170**”). The processor **160** executes various processes. Examples of the processor **160** include a general-purpose processor, a special-purpose processor, a central processing unit (CPU), a graphics processing unit (GPU), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), an integrated circuit, and/or combinations thereof. The memory device **170** stores various information. Examples of the memory device **170** include a volatile memory, a nonvolatile memory, a hard disk drive (HDD), and a solid state drive (SSD). The control device **150** may

include one or more electronic control units (ECUs). In general, the control device **150** may also be referred to as processing circuitry.

[0073] The vehicle control program PROG1 is a computer program executed by the processor **160**. The function of the control device **150** may be implemented by cooperation between the processor **160** that executes the vehicle control program PROG1 and the memory device **170**. The vehicle control program PROG1 is stored in the memory device **170**. Alternatively, the vehicle control program PROG1 may be recorded in computer-readable recording media.

4-2. Information on Sensor Detection

[0074] The control device **150** acquires the information on sensor detection SEN using the sensors **120**. The information on sensor detection SEN is stored in the memory device **170**. The information on sensor detection SEN includes the image IMG, vehicle status information, positional information, object information, and the like. The image IMG is captured by the camera CAM. The vehicle status information indicates the state (e.g., speed, steering angle, etc.) of the vehicle **100** detected by the vehicle state sensor. The positional information indicates the position and orientation of the vehicle **100** detected by the position sensor.

[0075] The object information is information about an object around the vehicle **100**. Examples of the object around the vehicle **100** include a walker, a bicycle, a two wheeled vehicle, other vehicle (a preceding vehicle, a parallel driving vehicle, a following vehicle, and the like), a white lane, road construction (e.g., a curbstone, a guard rail), a pole, a traffic light, a mark, and the like. The control device **150** may recognize an object around the vehicle **100** by using the recognition sensor. For example, by analyzing the image IMG, an object can be identified and the relative position of the object can be calculated. In addition, it is also possible to identify an object based on the point group information obtained by the LIDAR and acquire the relative position and the relative speed of the object. The object information includes a relative position of the object with respect to the vehicle **100**. The object information may further include a relative speed of the object.

4-3. Vehicle Travel Control

[0076] The control device **150** executes vehicle travel control for controlling the travel of the vehicle **100**. The vehicle travel control includes steering control, drive control, and braking control. The control device **150** may perform vehicle driving control by controlling the driving device **130** (e.g., a steering device, a driving device, and a braking device).

[0077] The control device **150** may perform autonomous driving control based on the information on sensor detection SEN. More specifically, the control device **150** generates a driving plan of the vehicle **100** based on the information on sensor detection SEN. Further, the control device **150** generates a target trajectory required for the vehicle **100** to travel in accordance with the travel plan based on the information on sensor detection SEN. The target trajectory includes a target position and a target speed. Then, the control device **150** performs vehicle travel control so that the vehicle **100** follows the target trajectory.

4-4. Processing Related to the Remote Support

[0078] When the remote support of the vehicle **100** is performed, the control device **150** communicates with the remote operator terminal **200** through the communication device **110**.

[0079] The control device **150** transmits at least a part of the information on sensor detection SEN to the remote operator terminal **200**. Typically, the control device **150** transmits the image IMG to the remote operator terminal **200**. The control device **150** may transmit the vehicle status information to the remote operator terminal **200**. The control device **150** may transmit the object information to the remote operator terminal **200**.

[0080] The control device **150** receives the remote support information OPE from the remote operator terminal **200**. The remote support information OPE is information on remote support by the remote operator O. For example, the remote support information OPE includes an operation amount by the remote operator O. The control device **150** performs the vehicle travel control in accordance with the received remote support information OPE.

4-5. Camera Information

[0081] The camera information CINF includes installation information and performance information on each of one or more camera CAMs mounted on the vehicle **100**. The installation information indicates the installation position and the installation orientation of the camera CAM in the vehicle coordinate system. The performance information indicates the focal length, the angle of view, and the like of the camera CAM. The camera information CINF is stored in the memory device **170**. The control device **150** may transmit the camera information CINF to the remote operator terminal **200**.

5. Configuration Example of the Remote Operator Terminal

[0082] FIG. **11** is a block diagram illustrating an example of a configuration of the remote operator terminal **200**. The remote operator terminal **200** includes a communication device **210**, a display device **220**, an input device **230**, and an information processing device **250**.

[0083] The communication device **210** may communicate with the vehicle **100** and the management device **300**.

[0084] The display device **220** displays various information on the remote operator O performing the remote support. In other words, the display device **220** presents the various information to the remote operator O by displaying the various information.

[0085] The input device **230** is a member operated by the remote operator O when the remote operator O remotely supports the vehicle **100**. For example, the input device **230** may include remote support members. Remote support members include a steering wheel, a gas pedal, a brake pedal, a turn signal, etc.

[0086] The information processing device **250** controls the remote operator terminal **200**. The information processing device **250** includes one or more processors **260** (hereinafter, simply referred to as a “processor **260**”) and one or more memory devices **270** (hereinafter, simply referred to as a “memory device **270**”). The processor **260** executes various processing. Examples of the processor **260** include a general purpose processor, an application specific processor, a CPU, a GPU, an ASIC, an FPGA, an integrated circuit, and/or combinations thereof. The memory device **270** stores various information. Examples of the memory device **270** include a volatile memory, a nonvolatile memory, an HDD, and an SSD. In general, the information processing device **250** may be referred to as processing circuitry.

[0087] The remote support control program PROG2 is a computer program executed by the processor **260**. The processor **260** that executes the remote support control program PROG2 and the memory device **270** may cooperate to implement the function of the information processing device **250**. The remote support control program PROG2 is stored in the memory device **270**. Alternatively, the remote support control program PROG2 may be recorded in computer-readable recording media. The remote support control program PROG2 may be provided via a network.

[0088] The information processing device **250** communicates with the vehicle **100** via the communication device **210**. The information processing device **250** receives the information on sensor detection SEN transmitted from the vehicle **100**. The information processing device **250** presents necessary information in the received information on sensor detection SEN to the remote operator O. For example, the information processing device **250** presents the image IMG to the remote operator O by displaying the image IMG on the display device **220**. The remote operator O can recognize the state of the vehicle **100** and the surrounding situation based on the presented information.

[0089] The remote operator O operates the input device **230**. The operation amount of the input device **230** is detected by a sensor installed in the input device **230**. The information processing device **250** generates remote support information OPE reflecting the operation amount of the input device **230** by the remote operator O. Then, the information processing device **250** transmits the remote support information OPE to the vehicle **100** via the communication device **210**.

[0090] The information processing device **250** may receive the camera information CINF

transmitted from the vehicle **100**. The camera information CINF is stored in the memory device **270**.

[0091] The information processing device **250** performs the projection transformation processing (the delay compensation processing and the movement adjustment processing) described in the above section 3 and the generation processing of the synthetic image described in the above section 2. The speed and steering angle of the vehicle **100** are obtained from information on sensor detection SEN. The steering angle in the steering operation by the remote operator O may be regarded as the steering angle of the vehicle **100**. The installation information and the performance information on each camera CAM mounted on the vehicle **100** are obtained from the camera information CINF. The information processing device **250** executes the projection transformation processing (the delay compensation processing and the movement adjustment processing) described in the above section 3 based on these pieces of information.

Claims

1. A device for remotely supporting a mobile vehicle, the device configured to remotely support the mobile vehicle by outputting a synthetic image in which annotation information about an object to be noted in an image is superimposed on the image obtained by a camera mounted on the mobile vehicle from a display device, the device comprising a communication circuitry connected to the mobile vehicle via a communication network; and a processing circuitry coupled to the communication circuitry, wherein the communication circuitry is configured to: receive, from the mobile vehicle, image data obtained by encoding the image and information on acquisition timing of the image by the camera; and receive, from the mobile vehicle, object data including recognition information on an object in the image separately from the image data, wherein the processing circuitry is configured to: set an image delay compensation time equal to or less than an image delay time indicating a timing difference between an acquisition timing of the image included in the image data and a decoding timing of the image data by the processing circuitry; perform projection transformation of the image included in the image data into a future image obtained at a camera viewpoint later than the acquisition timing of the image by the image delay compensation time based on information regarding a movement of the mobile vehicle; and generate the synthetic image based on the future image and recognition information on the object in an original image of the future image.

2. The remote support device according to claim 1, wherein the recognition information on the object includes information on an acquisition timing of the image used for recognizing the object, and wherein the processing circuitry is configured to set, as the image delay compensation time, a time equal to or longer than an object delay time indicating a timing difference between the acquisition timing of the image used for the recognition of the object and a reception timing of the object data by the processing circuitry.

3. The remote support device according to claim 1, wherein the recognition information on the object includes information on an acquisition timing of the image used for recognizing the object, and wherein the processing circuitry is configured to set, as the image delay compensation time, a time equal to or longer than a difference between the image delay time and an object delay time indicating a timing difference between a reception timing of the object data by the processing circuitry and the acquisition timing of the image used for recognition of the object.

4. The remote support device according to claim 1, wherein the recognition information on the object includes information on an acquisition timing of the image used for recognizing the object, and wherein the processing circuitry is further configured to: set an object movement adjustment time that is equal to or less than an object delay time indicating a timing difference between the acquisition timing of the image used for recognizing the object and a reception timing of the object data by the processing circuitry; when the object is a moving body, perform a projection

transformation in which a partial image of the moving body included in the image included in the image data and used for recognition of the moving body is transformed into a future partial image obtained at a camera viewpoint later than the acquisition timing of the image by the object movement adjustment time, based on information on relative movement of the moving body with respect to the mobile vehicle; and generate the synthetic image based on the future partial image, the future image, and the recognition information on the moving body in an original image of the future image.

5. The remote support device according to claim 4, wherein the processing circuitry is configured to set a time equal to the object delay time as the image delay compensation time.

6. A method for remotely supporting a mobile vehicle, the method including remotely supporting the mobile vehicle by outputting, from a display device, a synthetic image in which annotation information about an object to be noted in an image is superimposed on the image obtained by a camera mounted in the mobile vehicle, wherein the method includes: receiving image data in which the image and information on acquisition timing of the image by the camera are encoded from the mobile vehicle; receiving object data including recognition information on an object in the image from the mobile vehicle separately from the image data; setting an image delay compensation time equal to or less than an image delay time indicating a timing difference between a decoding timing of the image data and an acquisition timing of the image included in the image data; performing projection transformation of the image included in the image data into a future image obtained at a camera viewpoint later than the acquisition timing of the image by the image delay compensation time based on information regarding a movement of the mobile vehicle; and generating the synthetic image based on the future image and recognition information on the object in an original image of the future image.

7. A non-transitory computer-readable medium storing a program for remotely supporting a mobile vehicle, the program causing a computer to function to remotely support the mobile vehicle by outputting, from a display device, a synthetic image in which annotation information about an object to be noted in an image obtained by a camera mounted on the mobile vehicle is superimposed on the image, wherein the program is configured to cause the computer to execute: receiving image data in which the image and information on acquisition timing of the image by the camera are encoded from the mobile vehicle; receiving object data including recognition information on an object in the image from the mobile vehicle separately from the image data; setting an image delay compensation time equal to or less than an image delay time indicating a timing difference between a decode timing of the image data and an acquisition timing of the image included in the image data; performing projection transforming the image included in the image data into a future image obtained at a camera viewpoint later than the acquisition timing of the image by the image delay compensation time based on information regarding a movement of the mobile vehicle; and generating the synthetic image based on the future image and recognition information on the object in an original image of the future image.
